

AKI in a Kidney-Transplant Recipient

Differential = transplant-specific causes + the usual pre/intra/post-renal causes · let timing-from-transplant narrow it · toolbox: renal duplex US, BK/CMV PCR, CNI trough, biopsy + DSAs

5 transplant-specific causes (don't drop the usual ones)

Compartment	Transplant-specific
Pre-renal	Acute CNI toxicity (afferent + efferent vasoconstriction); renal artery stenosis / artery-or-vein thrombosis
Intra-renal	Acute rejection (cellular / antibody); viral (BK, CMV, adenovirus); chronic CNI toxicity; recurrent or de novo GN
Post-renal	Ureteral stenosis; extrinsic compression — lymphocele, perinephric hematoma, urinoma

Usual causes still apply: volume depletion, NSAIDs/ACEi, sepsis, ATN, AIN, contrast, obstruction (calculi/clot/cancer). Transplant causes are **added**, not substituted.

Timing from transplant narrows the differential

When	Think first
< 24 h	Hyperacute rejection (pre-formed DSAs); vascular thrombosis
First 3 mo	Acute CNI toxicity; acute rejection (cellular/humoral)
3 mo – 2 yr	Renal artery stenosis; BK nephropathy (peak ~10–13 mo); CMV nephritis (>3 mo)
Any time	Late rejection (esp. non-adherence); ureteral stenosis; usual AKI causes

Pattern recognition

Entity	Clues	Diagnose with
Acute CNI tox	Tremor/neurotox, HTN, ↑K, ↑uric acid	↑ CNI trough (usually)
Chronic CNI tox	Gradual asx ↑Cr; normal trough doesn't exclude	Biopsy (sclerosis, fibrosis)
Rejection	Graft pain, fever, HTN, proteinuria >1 g/d, hematuria	Biopsy; + DSAs (± dd-cfDNA)
BK nephropathy	Often asx; hemorrhagic cystitis	Plasma BK PCR ; urine PCR; biopsy
CMV nephritis	D+/R– highest risk; fever, ↓WBC/plt	CMV PCR ± biopsy
RAS / thrombosis	HTN, edema; vascular timing	Renal duplex US
Obstruction	↓UOP, suprapubic pain	Renal US (hydronephrosis)

The toolbox

- **Renal duplex US** — hydronephrosis (post-renal) + arterial flow (RAS/thrombosis)
- **Viral studies** — plasma + urine BK PCR, CMV PCR
- **CNI trough** — supratherapeutic supports acute toxicity
- **Biopsy + DSAs** (± donor-derived cell-free DNA) — the standard for rejection & chronic CNI tox
- Involve **transplant nephrology early**

Do / AVOID

- **DO** work the full pre/intra/post differential before reaching for transplant-specific tests
- **DO** review meds — azole antifungals (e.g. fluconazole) raise CNI levels → acute toxicity
- **DO** send **plasma** BK PCR (preferred) alongside urine; reduce immunosuppression for sustained DNAemia
- **AVOID** trusting a normal CNI trough to rule out chronic CNI toxicity
- **AVOID** attributing all dysfunction to rejection — biopsy + DSAs confirm before escalating immunosuppression

2024 update: plasma BK PCR screening (monthly to ~9 mo, then q3mo to 2 yr; 2nd Int'l Consensus, *Transplantation* 2024). dd-cfDNA helps exclude antibody-mediated rejection but does not replace biopsy (ASTS, Oct 2024).